



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : III Semester B.Tech IT
Course Code & Title : CD3291 & Data Structures and Algorithms
Name of the Faculty member : Mrs. G. Mareeswari, AP / IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Sorting and Searching / Hashing
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Problem based Learning
- **Justification:**

Hashing is chosen for problem-based learning in Data Structures and Algorithms (DSA) because it provides an excellent opportunity for students to apply theoretical knowledge through hands-on problem solving. By implementing hash functions and exploring various collision resolution techniques such as linear probing, quadratic probing, and double hashing, students gain practical experience in addressing real-world data storage and retrieval challenges. This approach helps them understand how different probing strategies affect performance factors like clustering, load factor, and search efficiency. Through problem-based exercises, learners enhance their logical reasoning, analytical thinking, and creativity in designing efficient hash-based solutions, thereby strengthening their core understanding of hashing concepts and their applications in DSA.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

In the previous class, the formulas for calculating the hash function using separate chaining, linear probing, quadratic probing and double hashing are discussed to the students. Several problems are shared with the students in the class, the formula and steps were revised once again by the instructor and the students were supposed to solve the given problem with their teammates. Finally, at the end of the session, the students were allowed to discuss their answers with the other team members. Some of the sample screenshots of students solved copy is shown in the Fig1. a and Fig1. b.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO3	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	2	2	2
Justification for correlation	Apply the knowledge of mathematical formulation for using hash function	Able to identify engineering problems with hash data structures	Able to design solutions for complex engineering problems using hash data structure	Able to solve problem individually or in team and using hash data structures	Able to explain the way to solve the problem using hash table with their team members.	Able to apply hash data structures to solve some computational problem.

- **Images / Screenshot of the practice:**

DSA

Karthik M
953624205020

2. Represent the following keys in a hash table with 2 slots with the hash function $H(x) = x \bmod 7$
 59, 700, 76, 85, 92, 73 and 101.

Resolve the collision using the following

- Separate Chaining
- Linear probing
- Quadratic probing

(i) Separate Chaining

(a) Insert (50)
 $H(50) = 50 \% 7 = 1$

(i) Insert (700)
 $H(700) = 700 \% 7 = 0$

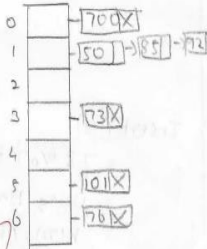
(ii) Insert (76)
 $H(76) = 76 \% 7 = 6$

(iv) Insert (85)
 $H(85) = 85 \% 7 = 1$

(v) Insert (92)
 $H(92) = 92 \% 7 = 1$

(vi) Insert (73)
 $H(73) = 73 \% 7 = 3$

(vii) Insert (101)
 $H(101) = 101 \% 7 = 5$



Data Structures and Algorithm

Abinaya . N
953624205001

1. Given Input (4371, 1323, 6173, 4199, 4344, 9679, 1989) by using Double Hashing method.
 Table size = 16

Insert 4371:

$4371 \% 10 = 1$ // No collision

Insert 1323:

$1323 \% 10 = 3$ // No collision

Insert 6173:

$6173 \% 10 = 3$ // collision.

$Hash_2 = 7 - (6173 \% 7)$
 $= 7 - 6 = 1$

$f(i) = 1 * 1$
 $= 1$ // collision

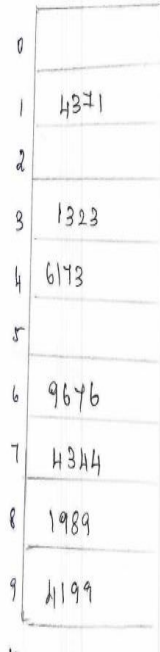


Fig 1.a Screenshot of problems solved by Karthik M and Abinaya N

K. Yamunadevi
953624205062.

DATA STRUCTURE AND ALGORITHM

Problem

1) Given input (4371, 1323, 6173, 4199, 4344, 9679, 1989) and a hash function,
 $h[x] = x \text{ mod } 10$. Do the following.
 (i) separate chaining
 (ii) linear probing
 (iii) quadratic probing

(i) Separate chaining

Insert, $4371 \div 10 = 1$

Insert, $1323 \div 10 = 3$

Insert $6173 \div 10 = 3$

Insert $4199 \div 10 = 9$

Insert $4344 \div 10 = 4$

Insert $9679 \div 10 = 9$

Insert $1989 \div 10 = 9$

(ii) Linear probing

Insert Key { 4371, 1323, 6173, 4199, 4344, 9679, 1989 }

953624205055 J.M. Thiraviyakumar

1) Given input (4371, 1323, 6173, 4199, 4344, 9679, 1989) hash function
 $h[x] = x \text{ mod } 10$

(i) Separate chaining

$4371 \div 10 = 1$
 $1323 \div 10 = 3$
 $6173 \div 10 = 3$
 $4199 \div 10 = 9$
 $4344 \div 10 = 4$
 $9679 \div 10 = 9$
 $1989 \div 10 = 9$

(ii) Linear probing

$4371 \div 10 \rightarrow h=1 \rightarrow$ Slot 1 empty $\rightarrow$ put at 1
 $1323 \div 10 \rightarrow h=3 \rightarrow$ Slot 3 empty $\rightarrow$ put at 3
 $6173 \div 10 \rightarrow h=3 \rightarrow$ Slot 3 occupied $\rightarrow$ check Slot 4 (occupied by 4344)
 $4199 \div 10 \rightarrow h=9 \rightarrow$ put at Slot 9
 $4344 \div 10 \rightarrow h=4 \rightarrow$ put at Slot 4
 $9679 \div 10 \rightarrow h=9 \rightarrow$ Slot 9 occupied $\rightarrow$ move to slot 0 $\rightarrow$ empty put at 0
 $1989 \rightarrow h=9 \rightarrow$ Slot 9 occupied $\rightarrow$ check 0 is occupied $\rightarrow$ check 1 occupied
 check 2 empty $\rightarrow$ put at 2
 $6173 \rightarrow h=3 \rightarrow$ Slot 3 occupied $\rightarrow$ Slot 4 occupied
 check 5 empty put it in slot 5.

Fig 1.b Screenshot of problems solved by Yamunadevi R and Thiraviyakumar J M

❖ Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

Students showed a strong interest in solving the given problems and approached them with enthusiasm. They were curious to explore different hashing techniques and enjoyed experimenting with various solutions. Many students expressed that working on the problem enhanced their understanding of the topic and made the learning process more engaging.

❖ Benefit of the practice:

- The activity improved students' practical understanding of hashing techniques through hands-on problem solving.

- It promoted teamwork, communication, and knowledge sharing among peers.
- Students developed analytical and critical thinking skills by exploring different probing methods and evaluating their efficiency.

❖ ***Challenges faced in implementation:***

- Some students initially struggled to understand the logic behind different probing techniques such as linear and quadratic probing.
- Slow learners required additional guidance to grasp the concept of collision resolution and implementation.

References:

- ❖ <https://www.hunschool.org/resources/problem-based-learning>
- ❖ <https://teaching.cornell.edu/teaching-resources/engaging-students/problem-based-learning>
- ❖ [https://citl.illinois.edu/citl-101/teaching-learning/resources/teaching-strategies/problem-based-learning-\(pbl\)](https://citl.illinois.edu/citl-101/teaching-learning/resources/teaching-strategies/problem-based-learning-(pbl))
- ❖ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3748308/>

Signature of the Faculty In-Charge

Signature of the HOD